Exercise 2
SQL Aggregate Function & operators.

(1) SElect Distinct department
FROM Students;

department
IT
HR
Finance

(02) Get the avarage age of Students per department

SELECT department, Avg(age) AS avg_age
From Students
group By department;

department	Avg_age
IT	20
HR	22
Finance	23

(03) Show department with more than 1 Student.

SELECT department, Count(*) AS Student_Count
FROM Student
Group By department
HAVING Count(*) > 1;

department	Student_Count
IT	2
HR	2

(04) get all students whose age is between 21 & 23.

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(07) List all courses that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;
```

Course-Id	Course-name	department	Credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

(08) Show the top 3 course by credits in descending order.

```
SELECT course_id, course_name, credits.
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

(09) Get the maximum, minimum, and avarage grade

```
SELECT MAX(grade) AS max_grade
       MIN(grade) AS min_grade
       AVG(grade) AS avg_grade
FROM enrollments;
```

Max_grade	Min_grade	Avg_grade
90	78	82

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(10) Count how many enrollments exits per course

```
SELECT course_id, count(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(11) Find total salary and total bonus per department

```
SELECT department, sum(salary) AS total_salary,
       sum(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

(12) Show department where average salary is above 55,000

```
SELECT department, avg(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary) > 55000;
```

department	Avg_Salary
IT	61000
Finance	70000

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(13) List employee whose salary plus bonus is greater than 60 000

```
SELECT employee_id, name, salary, bonus,
       (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

employee_id	Name	Salary	Bonus	total compensation
1	TOM	60000	5000	65000
3	SPIKE	70000	6000	76000
4	TYKE	62000	5500	67500

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